

AgriAdvisor

🌿 Track 5: Domain-Specialized AI Agents (Agriculture)

🔗 The Problem Statement

Track 5 - Domain-Specialized AI Agents (Agriculture): Agricultural advisory agents that work with multi-modal inputs (soil data, weather, market prices, voice in local languages) to give actionable, personalized guidance to farmers, even in low-connectivity environments.

Farmers in rural areas face a critical lack of timely, accurate, and localized agronomic advice. Low literacy rates, regional language barriers, and intermittent internet connectivity heavily restrict access to crucial insights regarding unpredictable weather changes, pest control, and volatile market pricing. Existing generic AI models fail to provide actionable, locally-grounded agricultural data when connectivity drops. Furthermore, the ratio of agricultural extension workers to farmers is extremely low, meaning personalized, on-demand guidance is virtually non-existent, leading to suboptimal crop yields, financial loss, and unchecked soil degradation.

💡 Our Solution: AgriAdvisor

We developed a highly resilient, multilingual mobile advisory platform powered by a hybrid AI architecture designed specifically for the agricultural domain.

- ✓ **Voice-First Interface:** Native speech-to-text and text-to-speech overcomes literacy and typing barriers, allowing farmers to converse naturally in their native dialects.
- ✓ **Multilingual Normalization:** Translates regional queries into a unified base language for the AI reasoning engine, then accurately localizes the generated response back to the farmer.
- ✓ **Resilient Hybrid Routing:** Dynamically utilizes a powerful **Express & LangGraph cloud pipeline** when connected, but gracefully degrades to **local Ollama inference models** (Llama3, Mistral) in zero-network zones to guarantee zero downtime.
- ✓ **Context-Aware Memory:** The app stores conversation context locally, allowing the AI to remember the farmer's specific crop cycle, soil type, and past issues without requiring repeated explanations.
- ✓ **Live Integrations:** Enriches answers with real-time Open-Meteo weather forecasting and live AGMARKNET Mandi prices, making the AI's advice highly actionable and economically relevant.

</> Core Technology Stack

📱 Frontend & App

- ✓ Expo 54 React Native
- ✓ TypeScript & Expo Router
- ✓ Native Voice (expo-av)
- ✓ Reanimated & Gestures
- ✓ TanStack React Query
- ✓ Zustand (State Mgt.)
- ✓ SQLite (Offline Cache)

🖥️ Backend & AI

- ✓ Express 5 & Node.js
- ✓ LangGraph Orchestrator
- ✓ Local Ollama (Llama3/Mistral)
- ✓ Python API Services
- ✓ JWT & bcryptjs Auth
- ✓ Pino (HTTP Logging)
- ✓ PM2 (Process Manager)

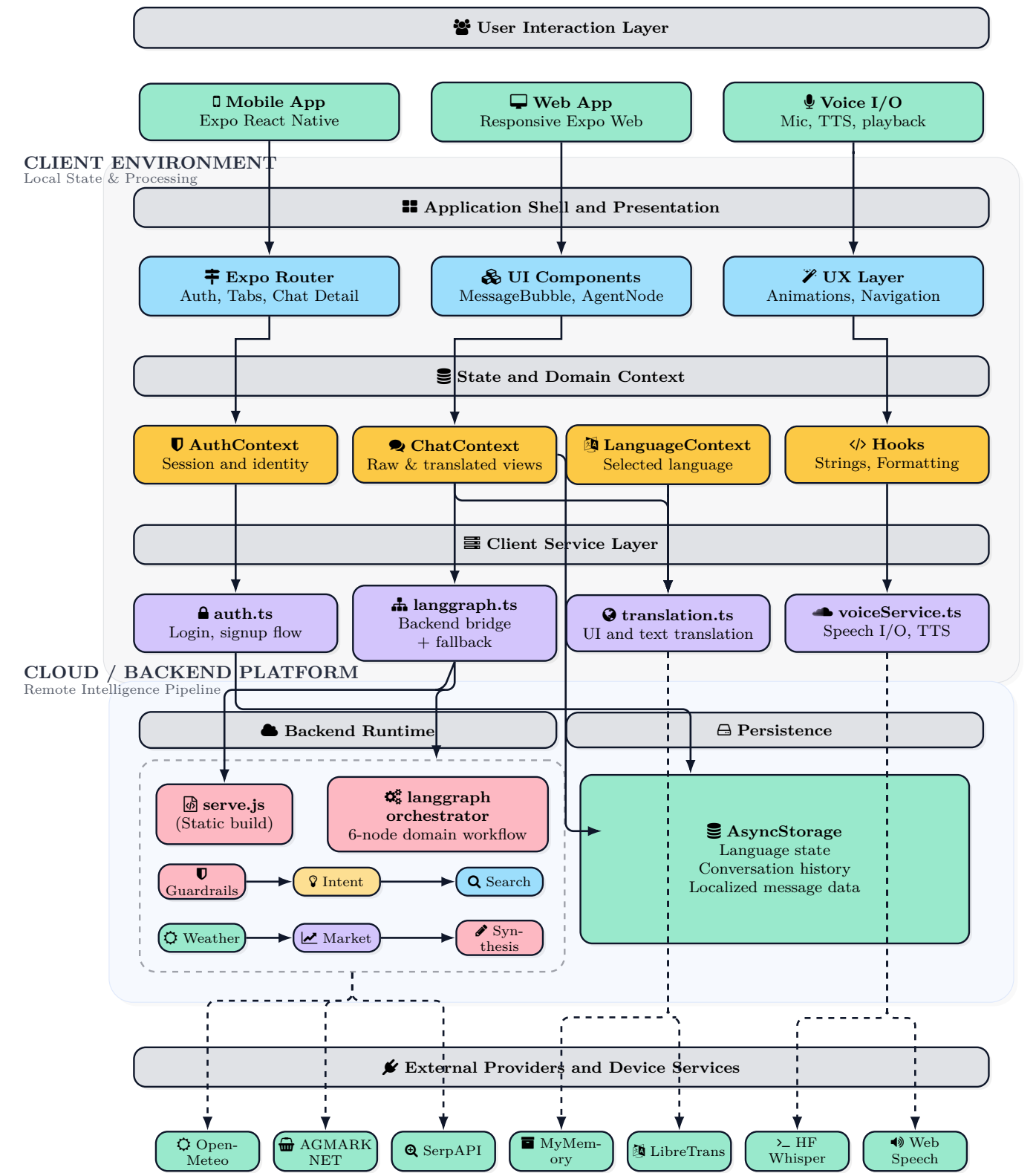
🗄️ Data & External APIs

- ✓ Neon Serverless Postgres
- ✓ Drizzle ORM & AsyncStorage
- ✓ Open-Meteo (Weather API)
- ✓ AGMARKNET (Live Prices)
- ✓ HF Whisper (Transcription)
- ✓ LibreTranslate (Localization)
- ✓ SerpAPI (Live Web Search)

AgriAdvisor Production Architecture

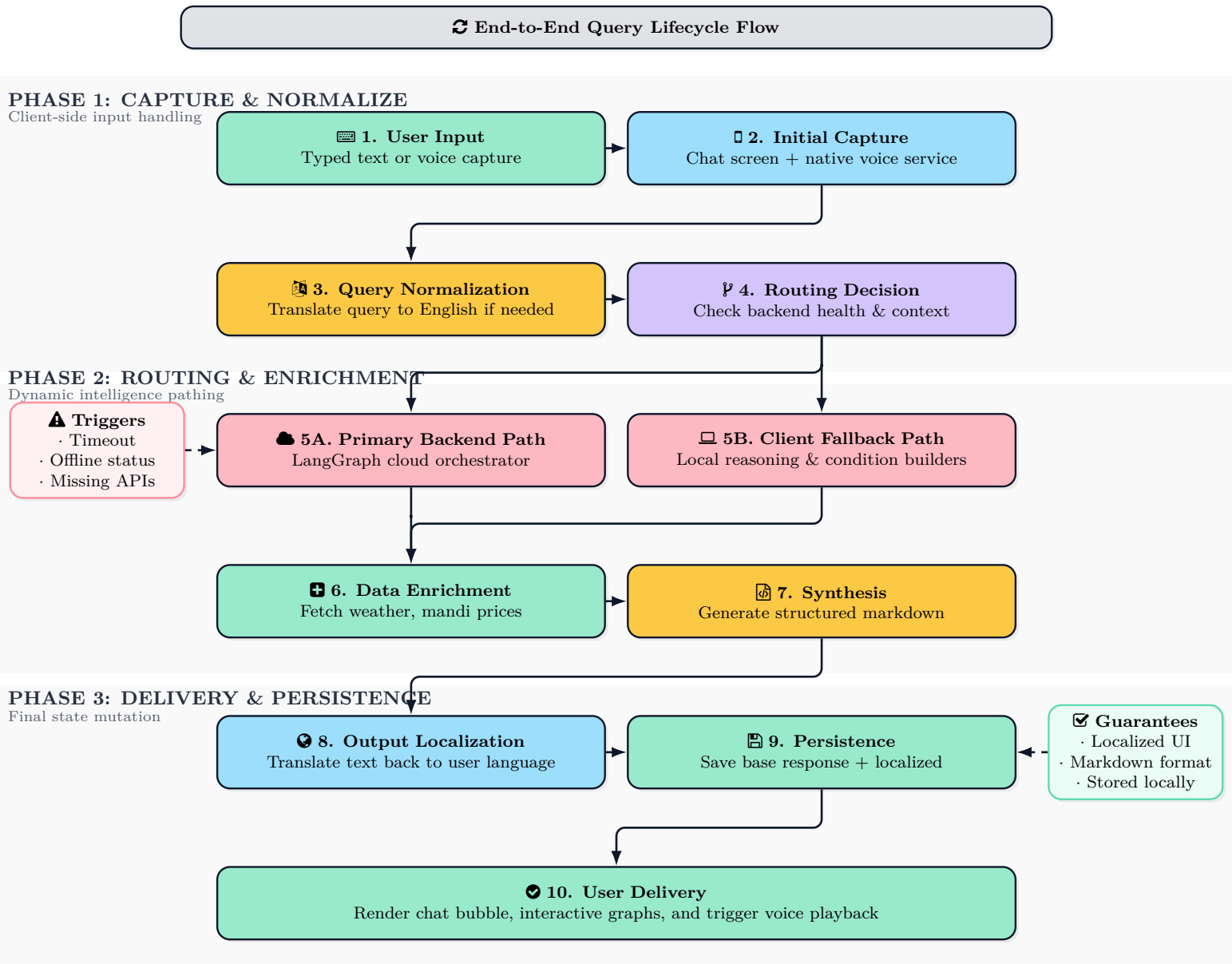
Multilingual mobile and web agricultural advisory platform with resilient orchestration and voice interfaces

Architecture objective. This design documents the current production-oriented architecture, capturing the runtime layers, state ownership, multilingual processing path, fallback behavior, data providers, and voice interfaces for mobile and web.



⇄ Runtime Request and Response Lifecycle

Execution model overview. Every user query initiates a resilient lifecycle designed to guarantee a localized, domain-accurate response. The system first captures input via text or native voice APIs, immediately normalizing the request into an English base for unified processing. Based on backend health and query intent, the orchestrator dynamically routes the request to either the primary LangGraph cloud pipeline or the local client fallback reasoning engine. After enriching the context with live external data, the system synthesizes a structured markdown response. Finally, this payload is translated back to the user’s selected language, persisted to local storage alongside the base response, and delivered through the UI and TTS engines.



📦 Component responsibility summary.

Layer	Primary responsibility	Implementation Context
📱 Presentation	Responsive mobile/web UI, chat rendering, graph formatting, voice controls	app/(tabs)/*, MessageBubble.tsx
🗄 State mgt.	Authentication state, language selection, conversation storage	AuthContext.tsx, ChatContext.tsx
📖 Client logic	Query normalization, backend routing, multilingual translation, speech	langgraph.ts, translation.ts
☁ Backend API	Domain-safe reasoning workflow with auditable node execution	langgraph_orchestrator.py
🔗 External APIs	Weather, market data, search, translation, Whisper transcription	Open-Meteo, AGMARKNET, Hugging Face

Project Conclusion & Roadmap

≡ Strategic Vision for AgriAdvisor

🚩 Conclusion

AgriAdvisor successfully demonstrates how modern AI can be engineered to bridge the digital divide in agriculture. By moving away from brittle, cloud-only dependencies, we have architected a system that respects the realities of rural infrastructure.

Through the strategic combination of **Expo's cross-platform capabilities**, **hybrid Edge/Cloud LangGraph orchestration**, and **local Ollama fallback models**, AgriAdvisor guarantees that critical agronomic intelligence is always accessible. The integration of native voice processing and real-time market/weather data transforms the AI from a simple chatbot into a highly capable, localized, and accessible digital extension worker. This platform lays a highly scalable foundation for empowering farmers, increasing crop yields, and preventing agricultural revenue loss.

🗺️ Future Scope & Roadmap

While the current architecture solves immediate communication and connectivity barriers, the platform is designed to scale horizontally. Our roadmap focuses on deepening the AI's sensory inputs and expanding its economic impact.

📷 Phase 1: On-Device Visual Diagnostics

We plan to integrate lightweight, on-device computer vision models (such as MobileNet v3) directly into the Expo mobile application. This will allow farmers to take photos of diseased leaves or pests and receive instant, offline disease identification and treatment plans without needing to upload high-resolution images over slow 2G/3G networks.

📶 Phase 2: IoT & Farm Sensor Integration

AgriAdvisor will be expanded to interface with low-cost Bluetooth/IoT agricultural sensors. By ingesting live telemetry data regarding soil moisture, NPK (Nitrogen, Phosphorus, Potassium) levels, and local ambient temperature, the LangGraph reasoning engine can generate hyper-personalized, precision-irrigation and fertilizer schedules.

🗣️ Phase 3: Hyper-Local Dialect Fine-Tuning

While current API translation handles major regional languages, deep rural pockets often use highly specific agricultural colloquialisms. We intend to fine-tune our Whisper transcription models and local LLM routing specifically on these micro-dialects to drastically improve speech-to-text accuracy and user trust.

📈 Phase 4: Predictive Yield & Economic Modeling

By analyzing a farmer's historical crop data alongside live AGMARKNET pricing trends and Open-Meteo long-term climate forecasts, the AI will upgrade from a reactive advisor to a proactive economic planner. It will be able to advise farmers on the most profitable harvest windows and alert them to market gluts before they occur.